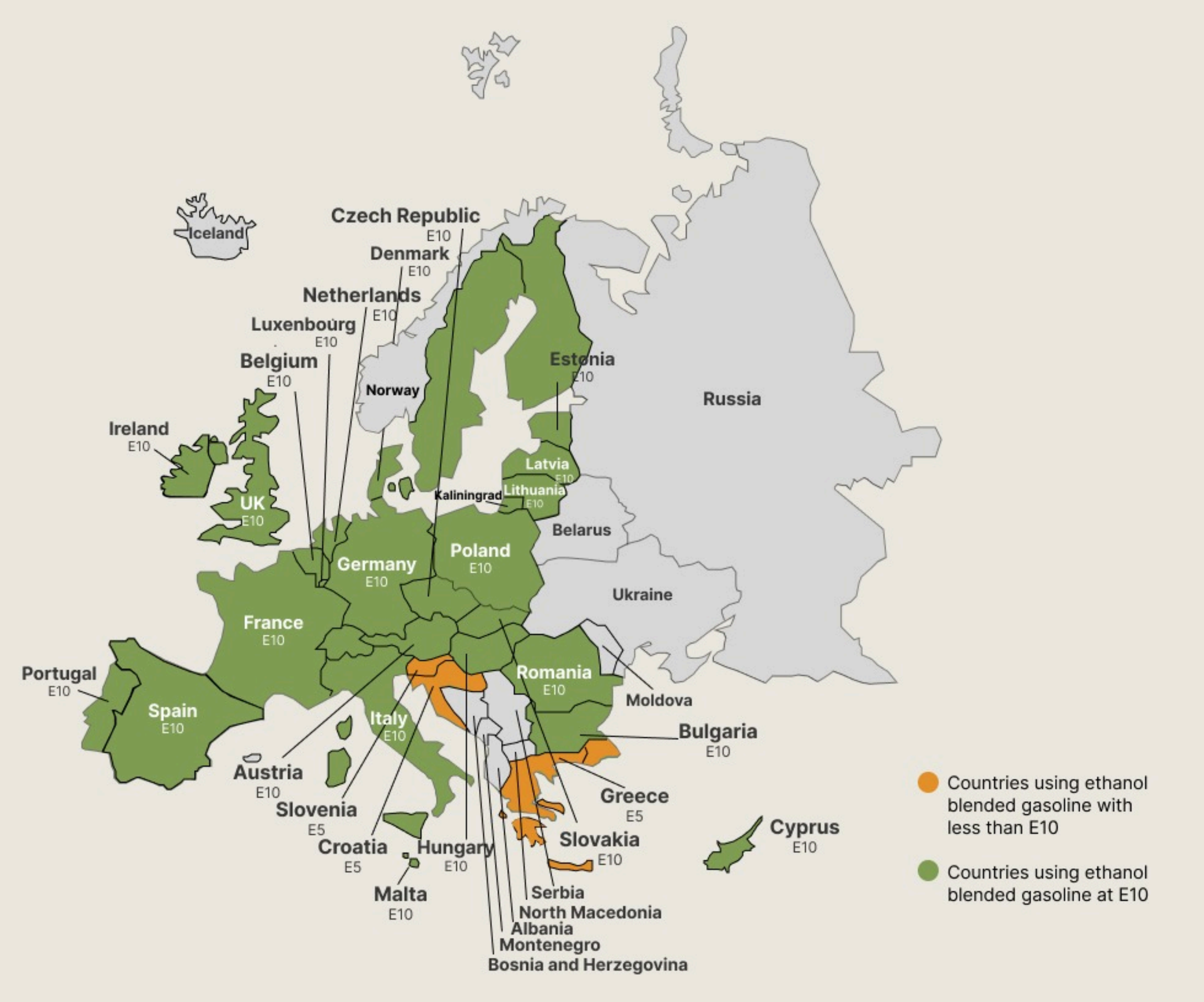
Ethanol Blending in Gasoline - Luxembourg

September, 2025

Ethanol Blending in Europe

There are important fuel quality and environmental impact of vehicle emission challanges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

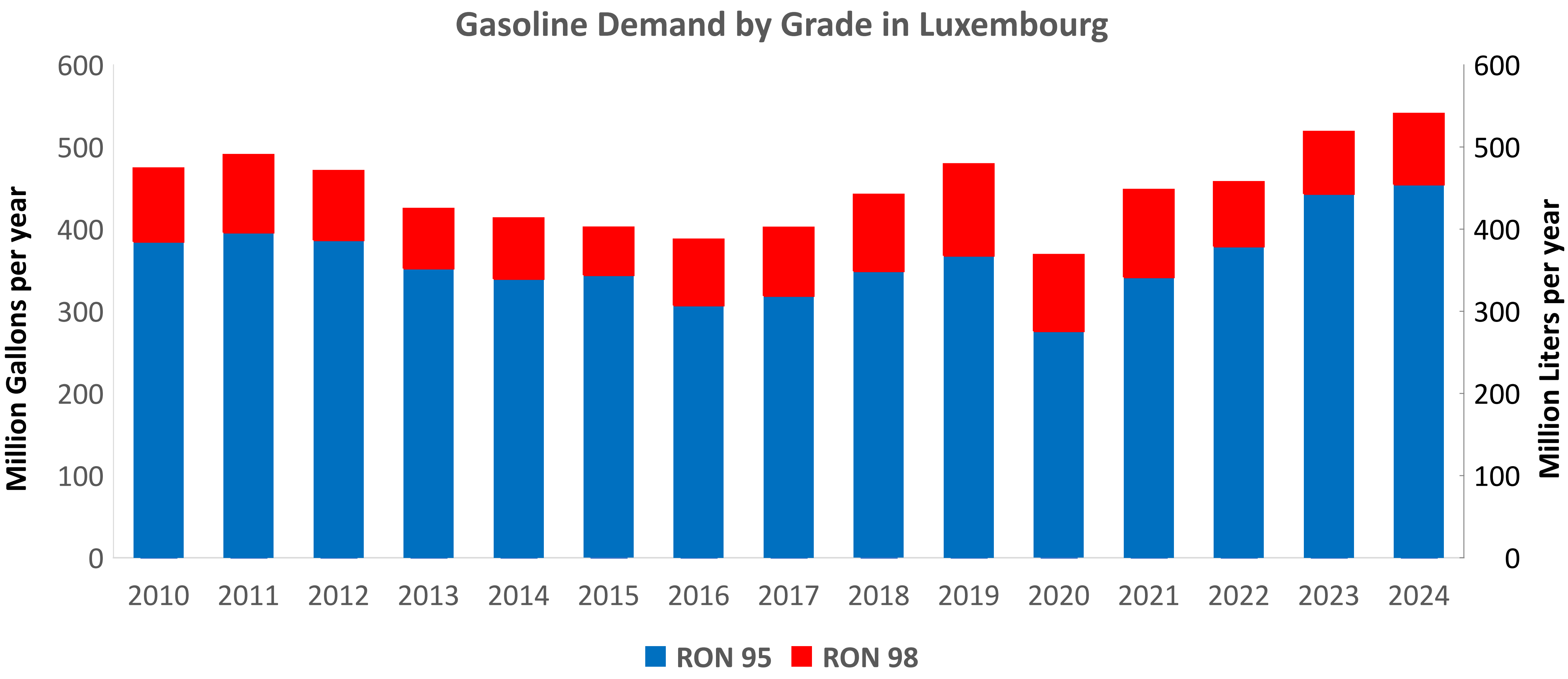
Ethanol Gasoline Blending in Luxembourg

In 2024, gasoline consumption in Luxembourg was 540 million liters (143 million gallons) and growth rate was 2.4%. There is no gasoline production. Gasoline net imports were 540 million liters (143 million gallons) correspondingly. Luxembourg complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.5 RON.

In 2024, ethanol consumption in Luxembourg was 29 million liters (8 million gallons) representing 5.3% of gasoline demand. Net Imports were 29 million liters (8 million gallons). Ethanol sources are Sugar Cane 80% and Corn 20%.

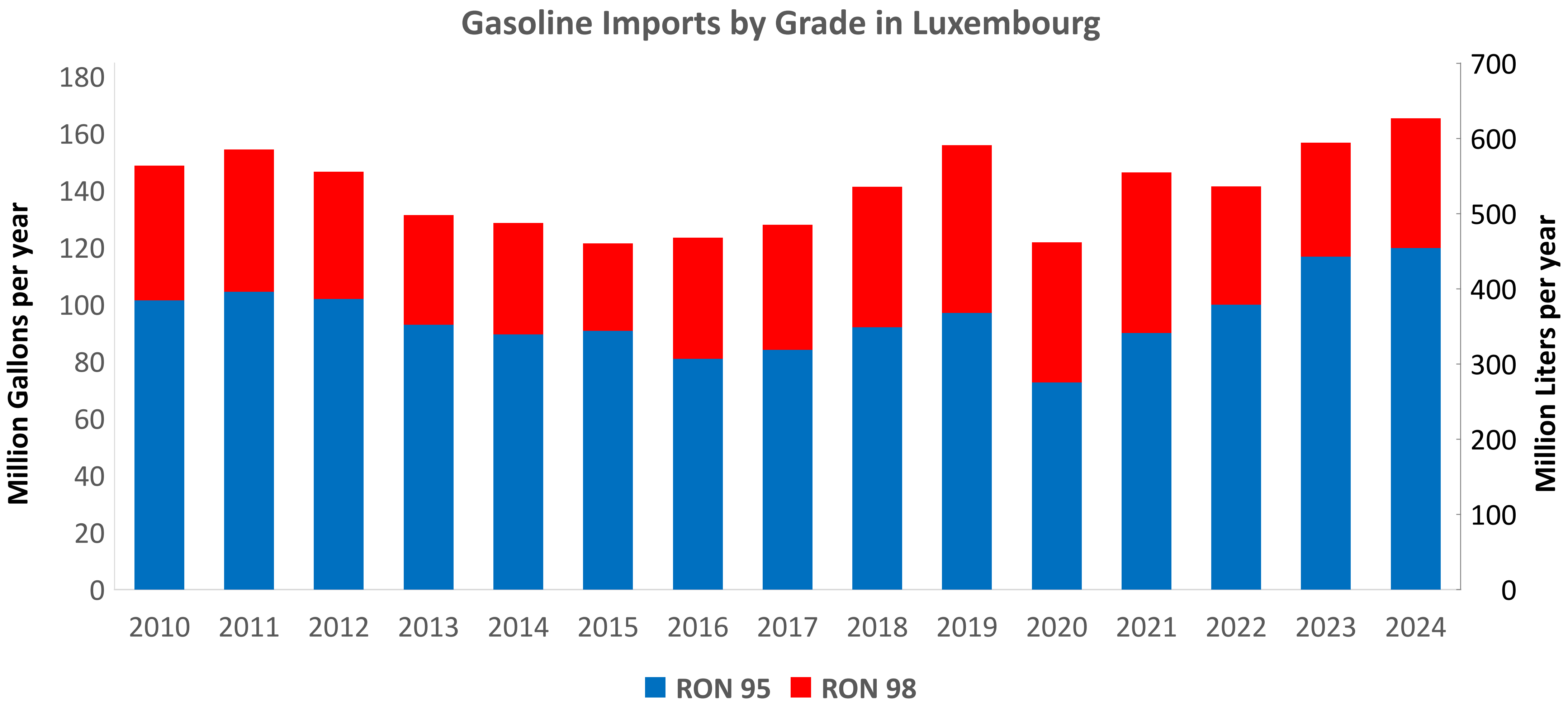
Source: Eurostat, European Commision

Gasoline Demand in Luxembourg



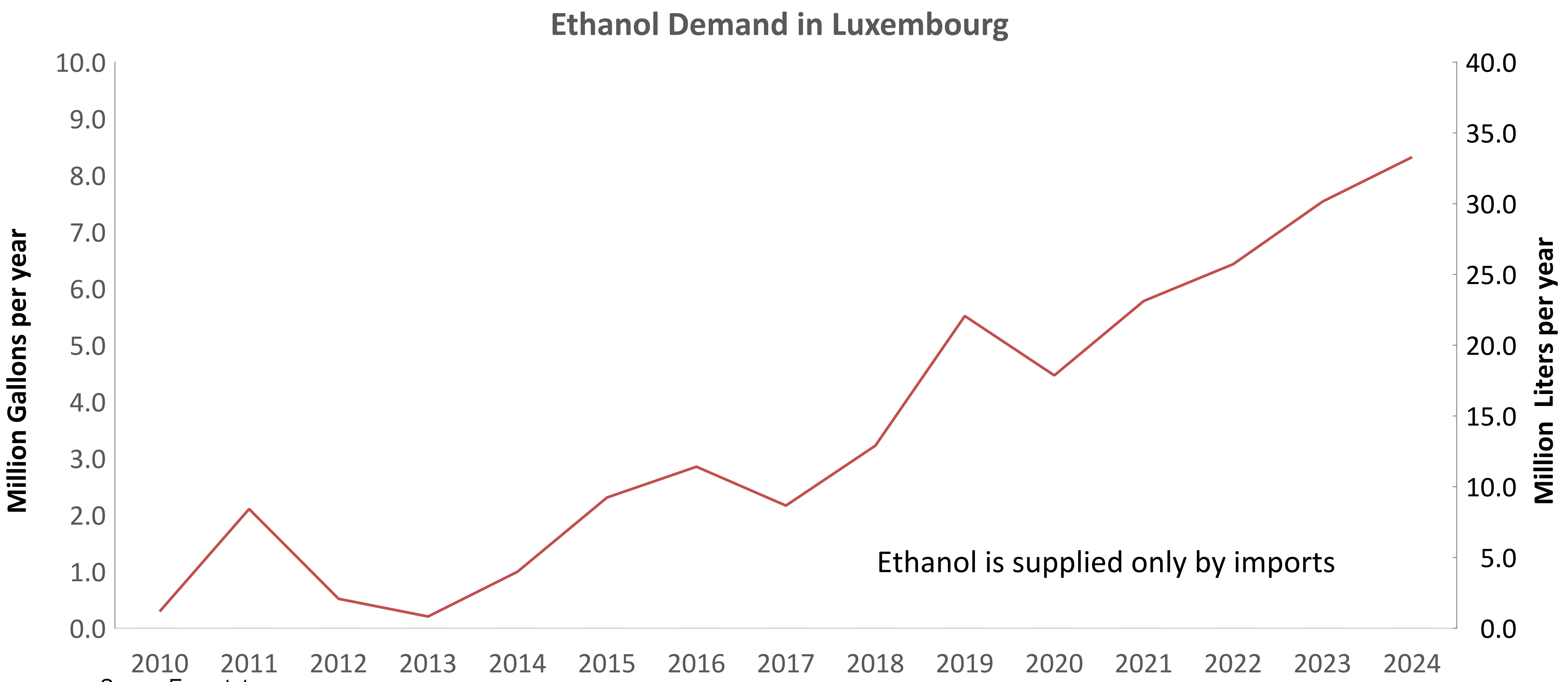
Source: Eurostat

Gasoline Imports in Luxembourg



Source: Eurostat

Ethanol Balance in Luxembourg



Source: Eurostat

Gasoline Quality in Luxembourg

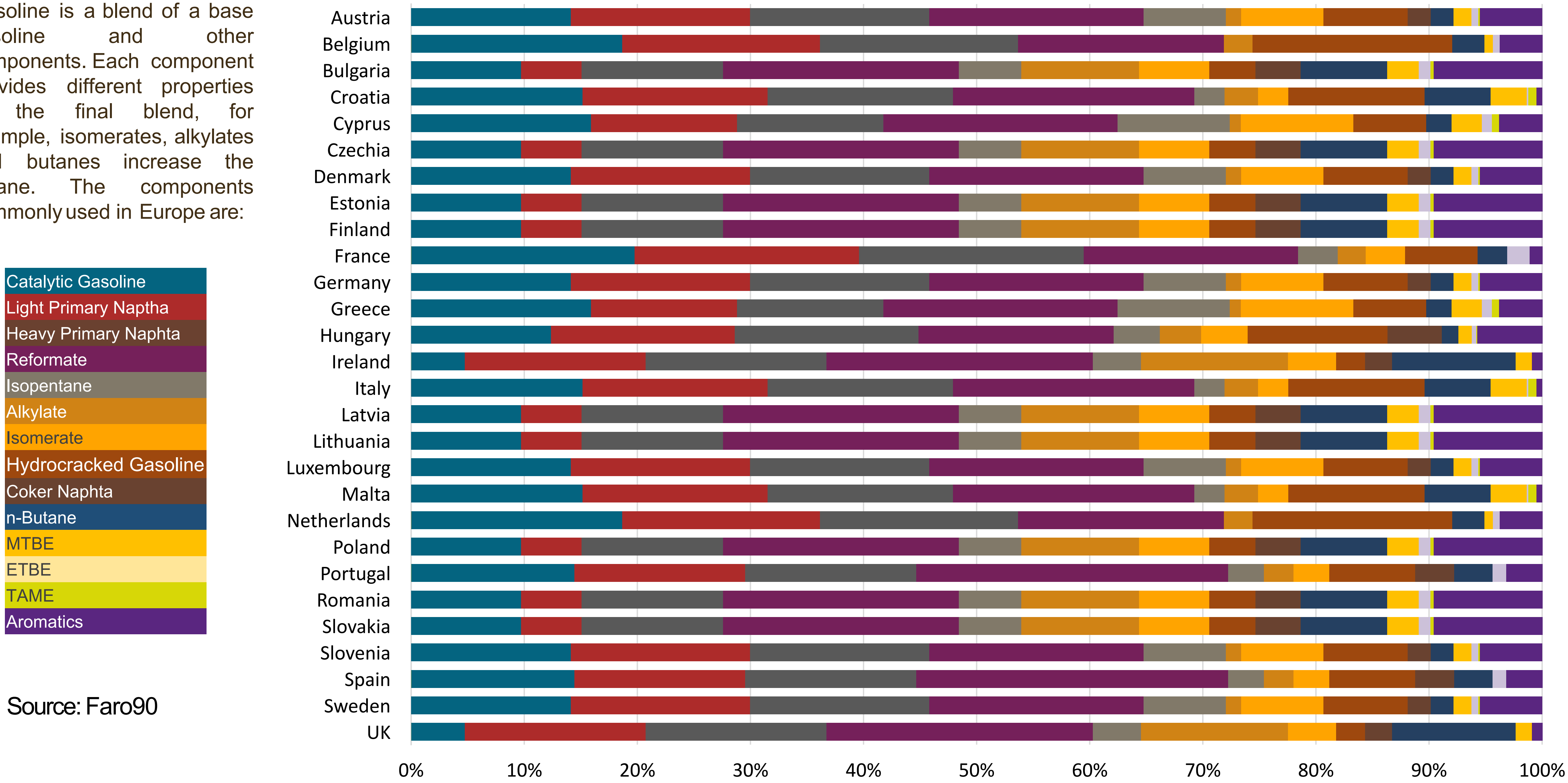
Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	7.7							
Advanced Biofuels (%cal)	-							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	6.0							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Gasoline Component Blending in Europe

Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:



Source: Faro90

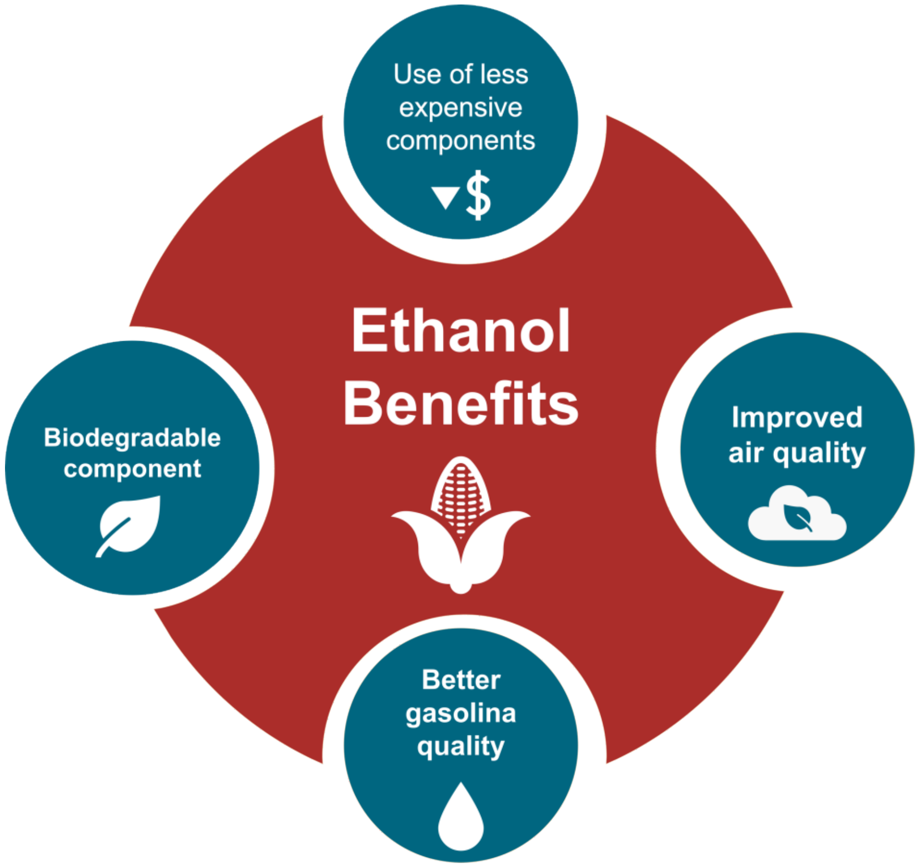
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component.The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives.To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

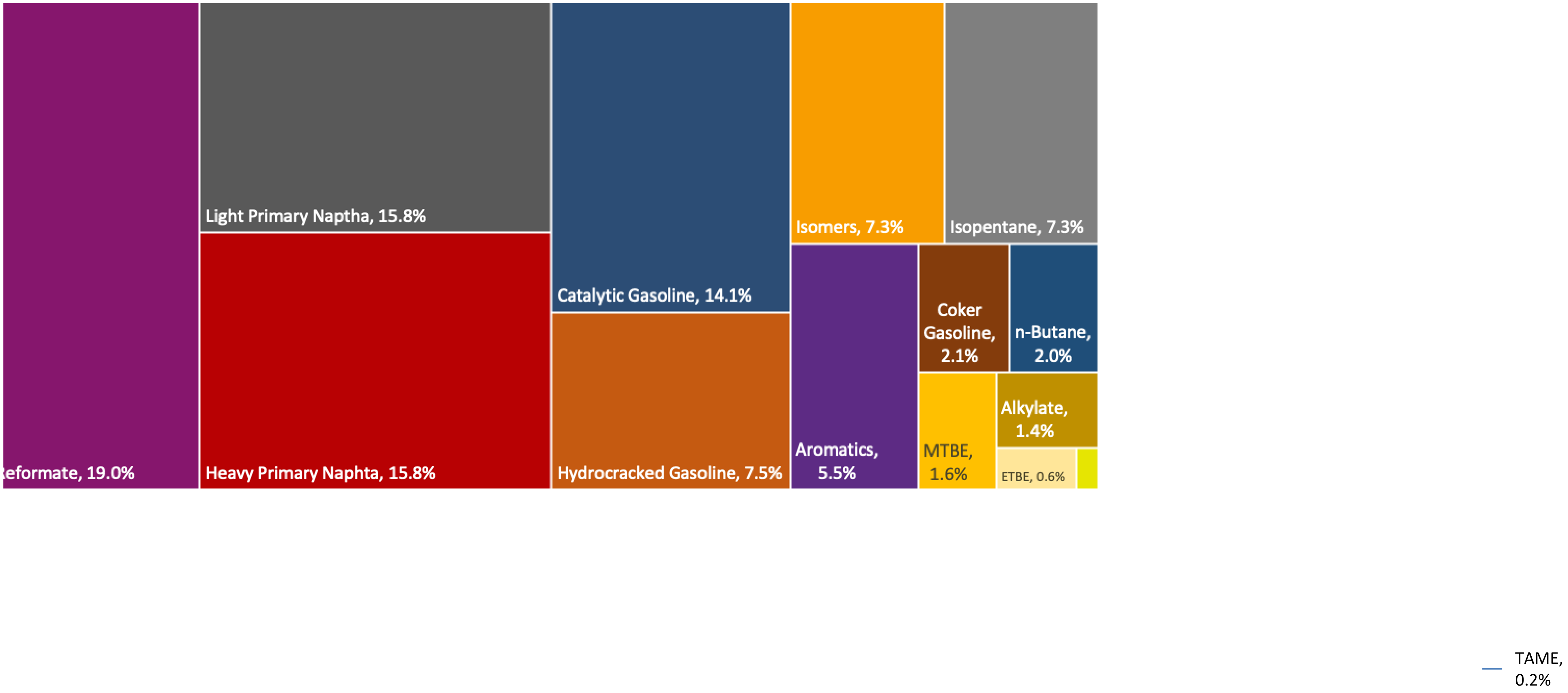
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

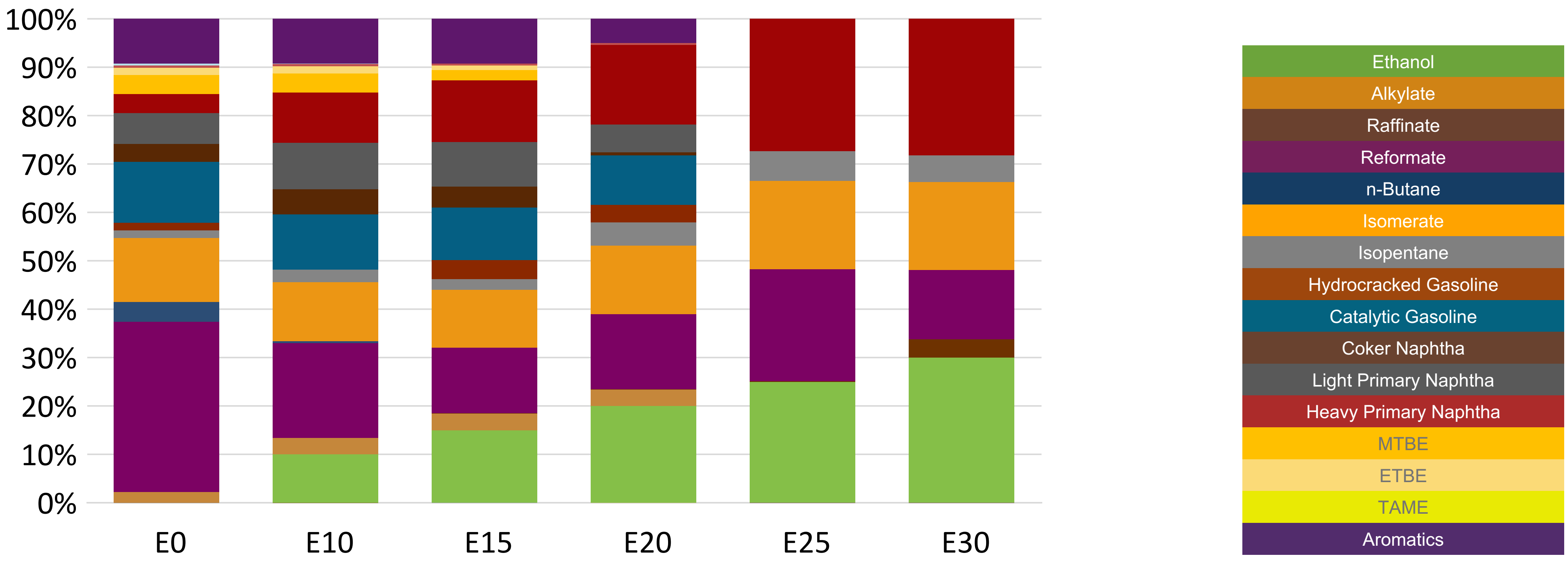


Luxembourg Available Blending Components



Source: Faro90

Luxembourg – Gasoline 95 – Constant Octane

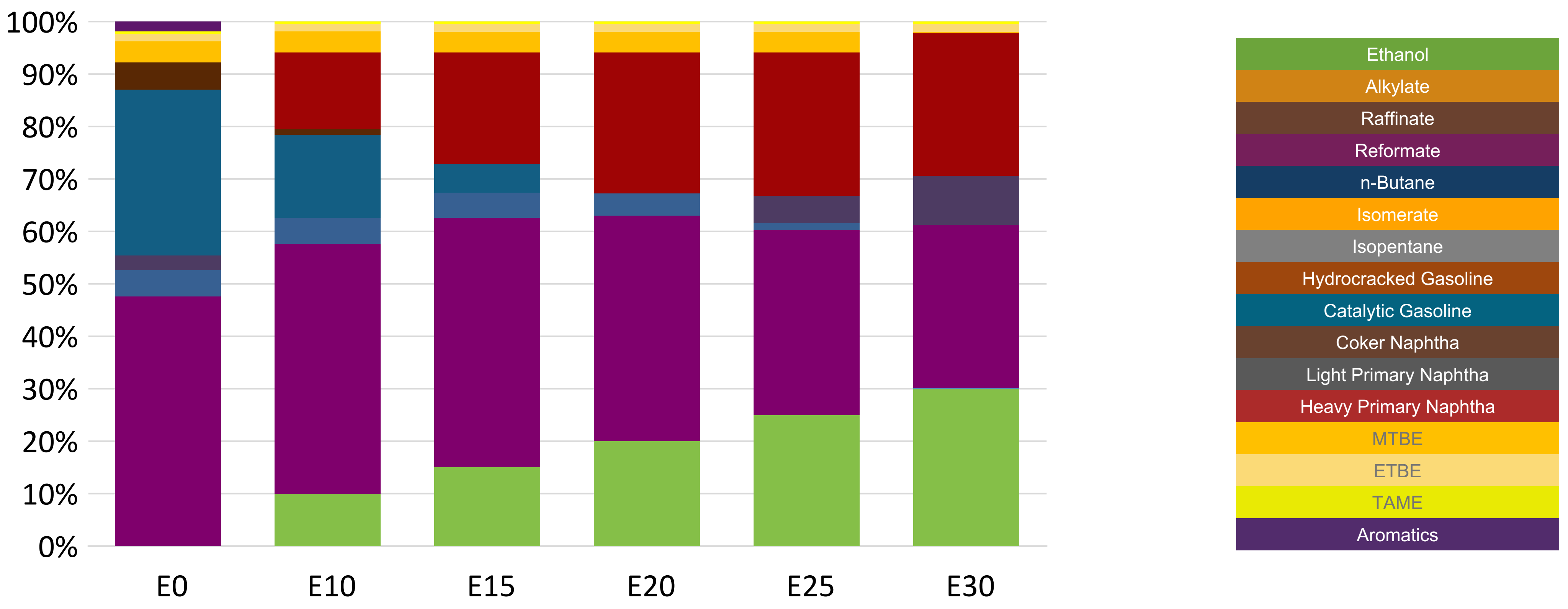


Octane (RON)	95.0	95.0	95.0	95.0	95.1	95.6
Price (US$/gal)	$ 2.272	$ 2.144	$ 2.083	$ 2.002	$ 1.880	$ 1.814

Average CIF 2024 Prices

Source: Faro90

Luxembourg - Gasoline 98 - Constant Octane

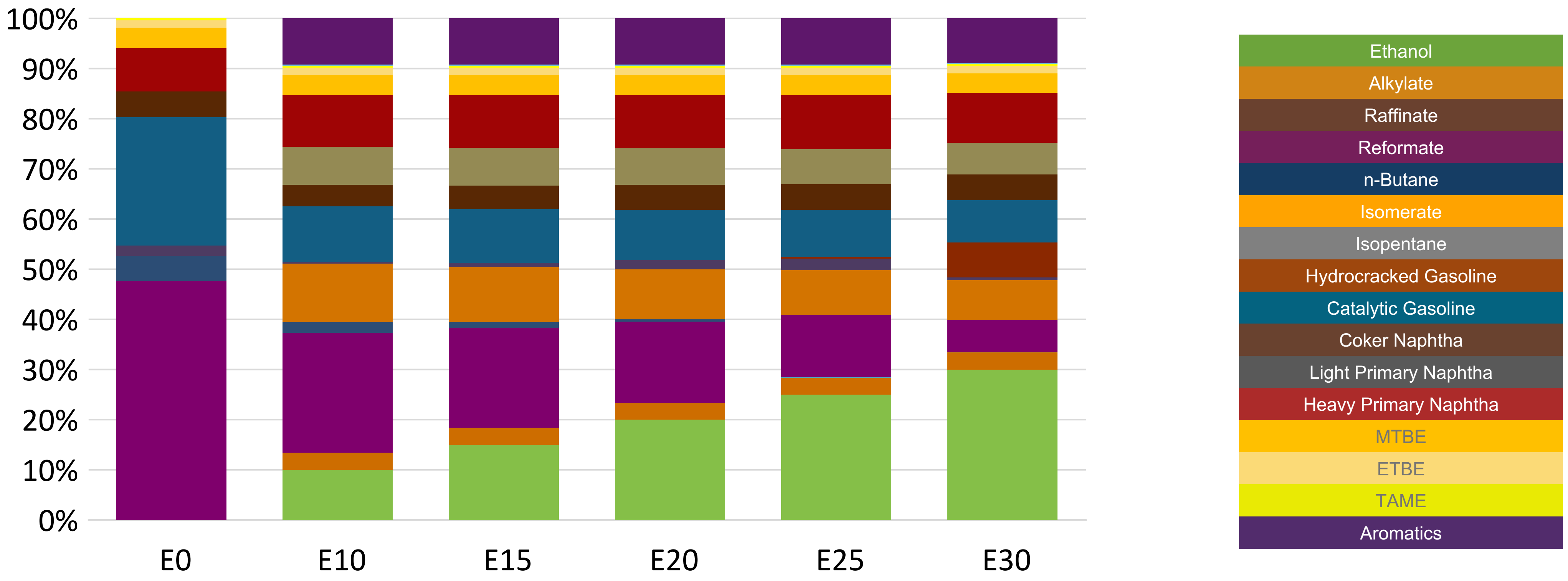


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US$/gal)	$ 2.347	$ 2.218	$ 2.146	$ 2.069	$ 1.992	$ 1.925

Average CIF 2024 Prices

Source: Faro90

Luxembourg – Gasoline 95 – Increased Octane

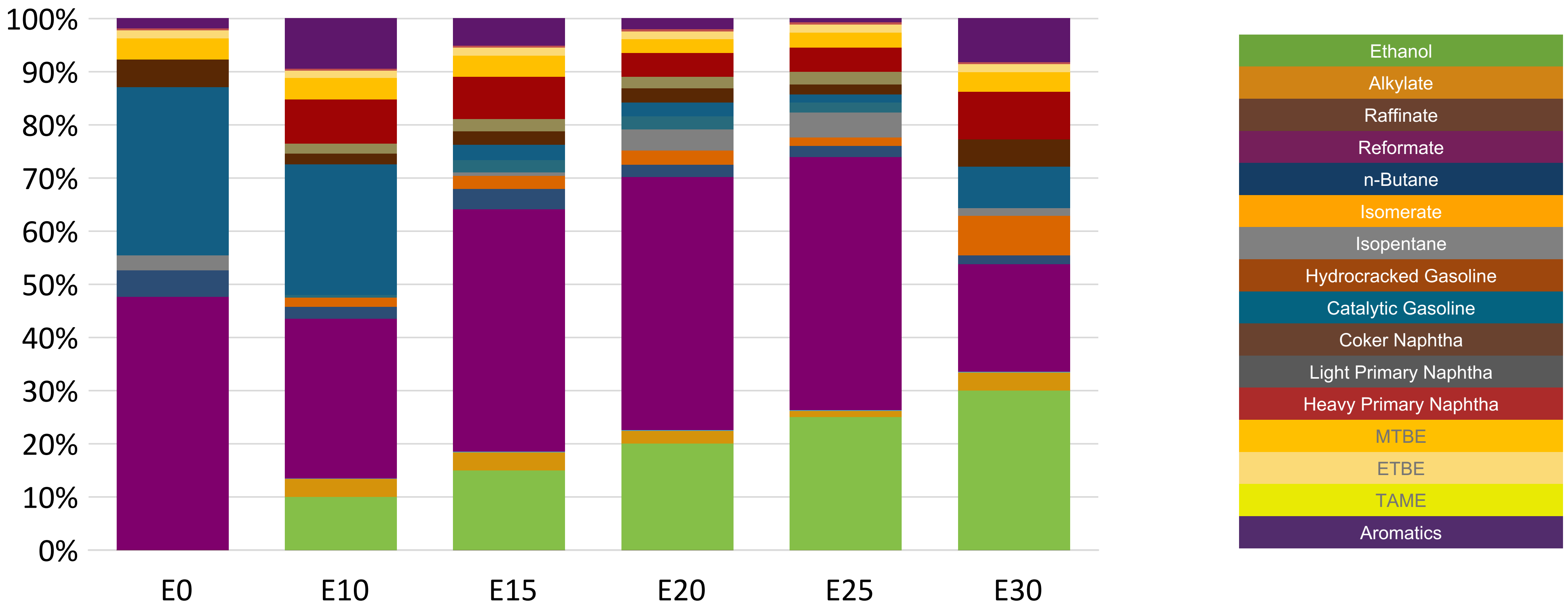


Octane (RON)	95.0	96.5	97.9	99.4	100.8	102.0
Price (US$/gal)	$ 2.209	$ 2.209	$ 2.209	$ 2.209	$ 2.209	$ 2.209

Average CIF 2024 Prices

Source: Faro90

Luxembourg – Gasoline 98 – Increased Octane



Octane (RON)	98.0	99.2	101.3	102.6	104.0	104.7
Price (US$/gal)	$ 2.347	$ 2.347	$ 2.347	$ 2.347	$ 2.347	$ 2.347

Average CIF 2024 Prices

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

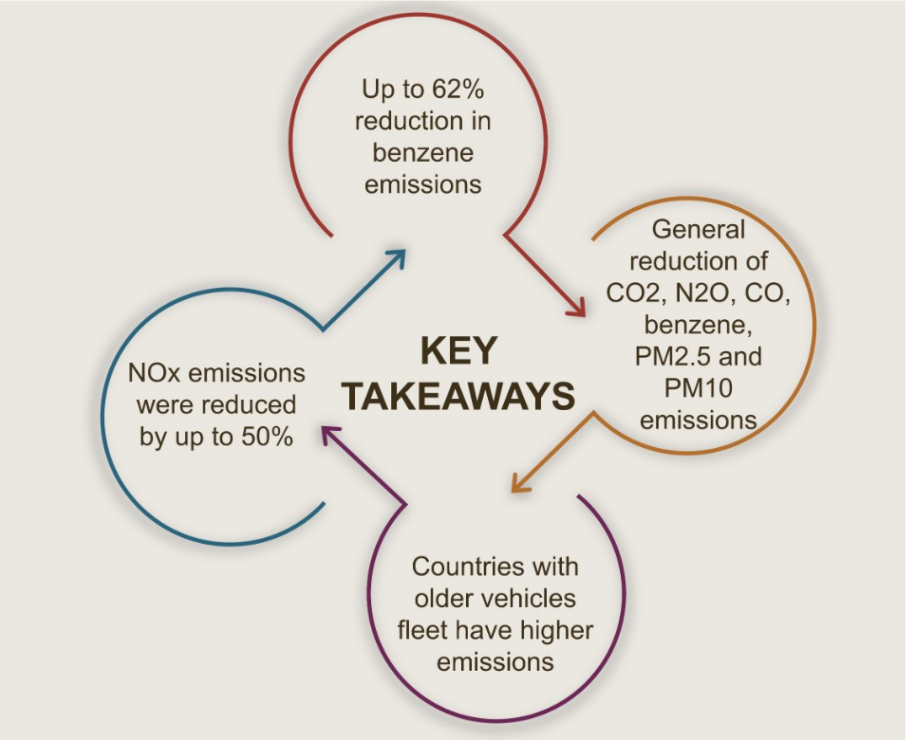
The model used in this analysis takes as a reference the **[International Vehicle Emissions Model (IVE).]**
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



Source: Bureau of transportation statistics.

Gasoline Vehicle Fleet - Luxembourg

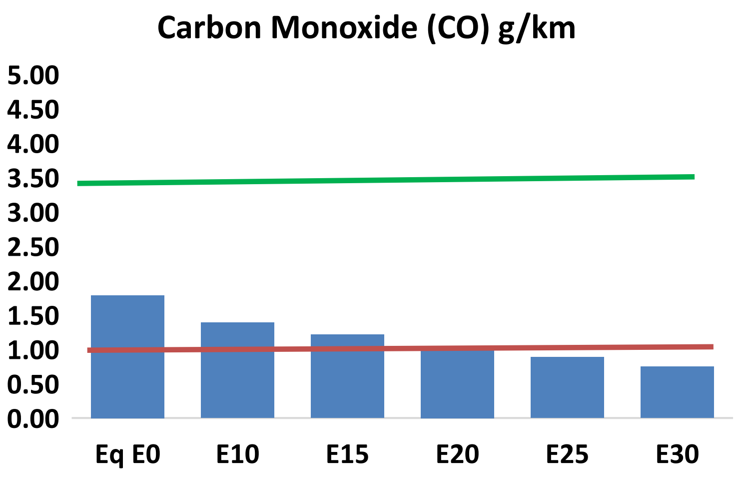
Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	1
		Single FI	3-Ways		>8 years	79
		Multi FI	3-Ways		>8 years	2,258
			Euro III		>8 years	12,158
			Euro IV		>8 years	94,410
			Euro V		>8 years	74,496
			Euro 6		>8 years	202,467
					4-8 years	256,114
					<4 years	144,161
			Hybrid		All	22,891
		Electric			All	20,545
	Medium	Carburator	3-Ways	PCV Tank	>8 years	0
		Single FI	3-Ways		>8 years	6
		Multi FI	3-Ways		>8 years	153
			Euro III		>8 years	797
			Euro IV		>8 years	5,974
			Euro V		>8 years	4,989
			Euro 6		>8 years	15,681
					4-8 years	25,748
					<4 years	25,672
			Hybrid		All	0
		Electric			All	474
Sml Engines	Light	4-Cycle	None	PCV	>8 years	114
			Euro III		>8 years	595
			Euro IV		>8 years	4,558
			Euro V		>8 years	3,628
			Euro 6		>8 years	11,903
					4-8 years	13,472
					<4 years	16,083
			Hybrid		All	0
		Electric			All	233
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	0
		Fuel-Injection	3-Ways		>8 years	36
			Euro III		>8 years	358
			Euro IV		>8 years	5,344
			Euro V		>8 years	5,786
			Euro 6		>8 years	16,314
					4-8 years	24,361
					<4 years	26,467
			Hybrid		All	6
		Electric			All	420

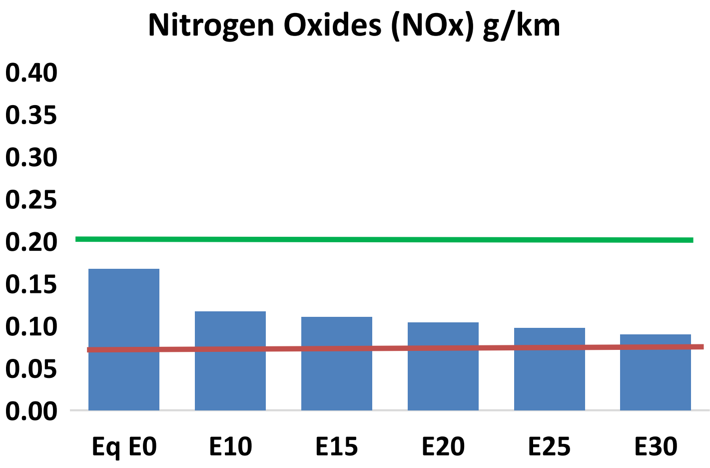
Vehicle Fleet: 1,038,755
Gasoline Vehicle Fleet: 694,397

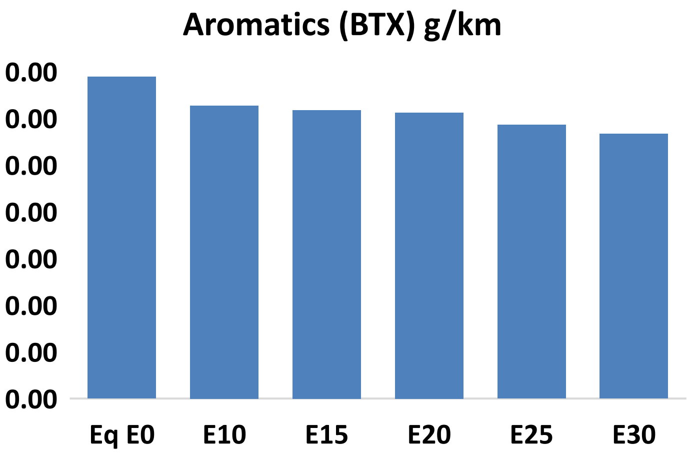
Source: Eurostat, ACEA

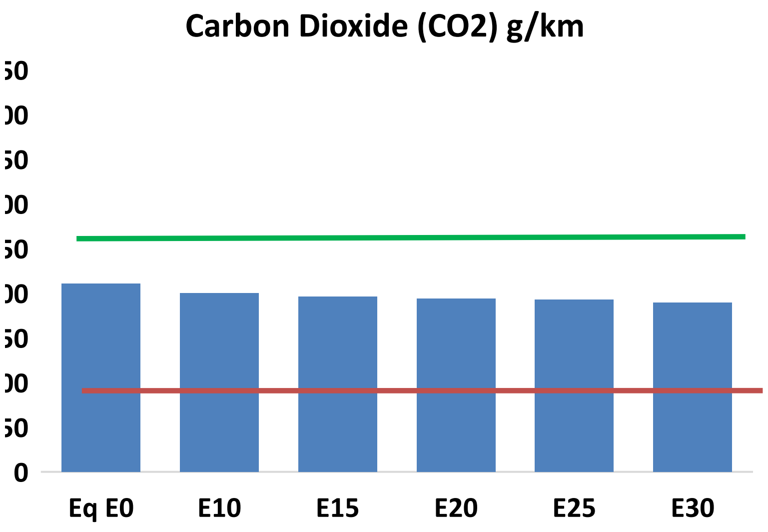
Luxembourg – Vehicular Emissions

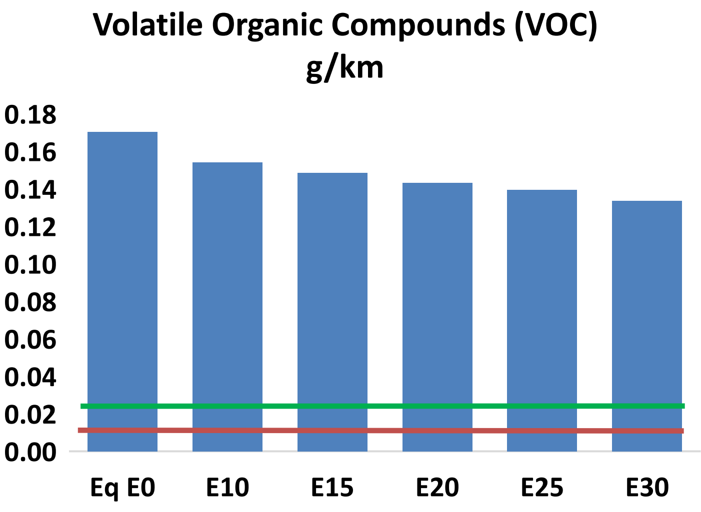
TIER III BIN 125 United States
Euro 6 Standard

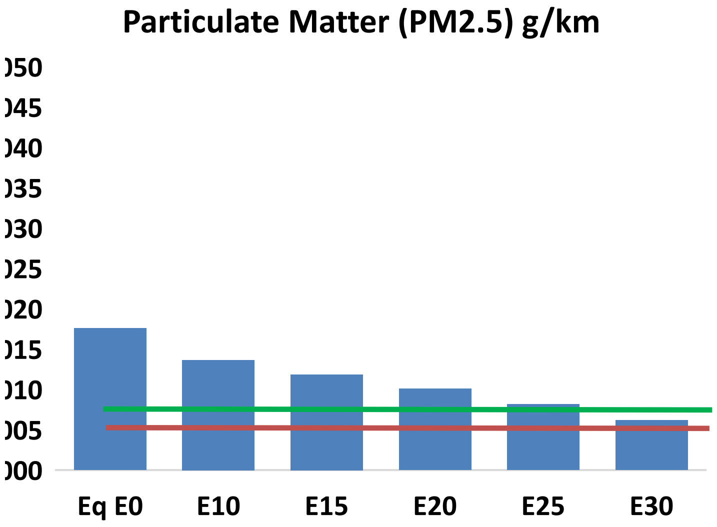












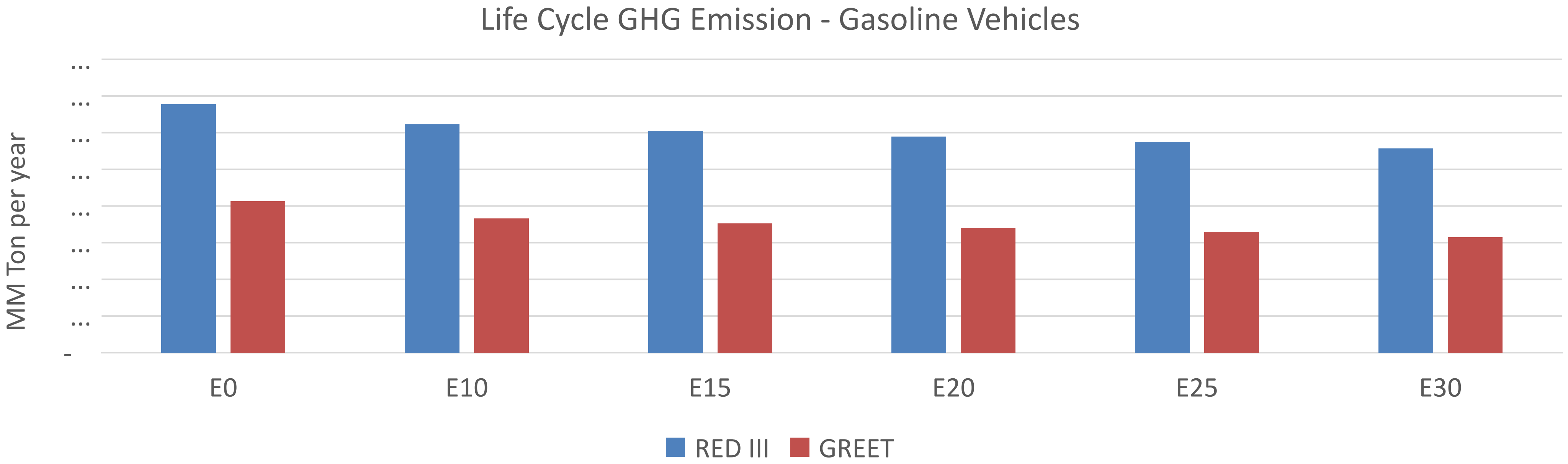
Source: Faro90

Luxembourg – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	32,931	29,249	25,567	22,326	18,961	16,342	13,726	-22%	-42%
CO2	3,882,918	3,783,071	3,688,772	3,614,608	3,577,909	3,556,182	3,490,631	-5%	-8%
VOC	3,142	2,992	2,843	2,738	2,642	2,570	2,465	-10%	-16%
NOx	3,081	2,311	2,157	2,033	1,917	1,790	1,655	-30%	-38%
SOx	44	41	37	34	32	29	26	-15%	-28%
NH3	1,305	1,292	1,279	1,285	1,299	1,309	1,318	-2%	0%
N2O	48	47	47	47	48	49	49	-1%	2%
CH4	747	747	747	758	775	789	802	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	31	29	28	27	27	26	25	-8%	-14%
Acetaldehyde	55	64	93	142	194	221	261	68%	249%
Formaldehyde	162	157	184	216	226	250	272	13%	39%
Benzene	63	61	58	57	56	54	52	-9%	-11%
PM2.5	244	216	189	164	139	113	86	-22%	-43%
PM10	81	72	62	54	46	37	28	-22%	-43%

Source: Faro90

Luxembourg – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	9.0
Target @ 2035 (MMT/year)	5.1
Delta	3.9

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	11.2%	14.7%	17.5%	20.2%	23.8%
RED II/III	0.0%	8.1%	10.8%	13.0%	15.2%	17.8%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	5.9%	7.7%	9.2%	10.6%	12.5%
RED II/III	0.0%	7.0%	9.3%	11.2%	13.1%	15.4%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90